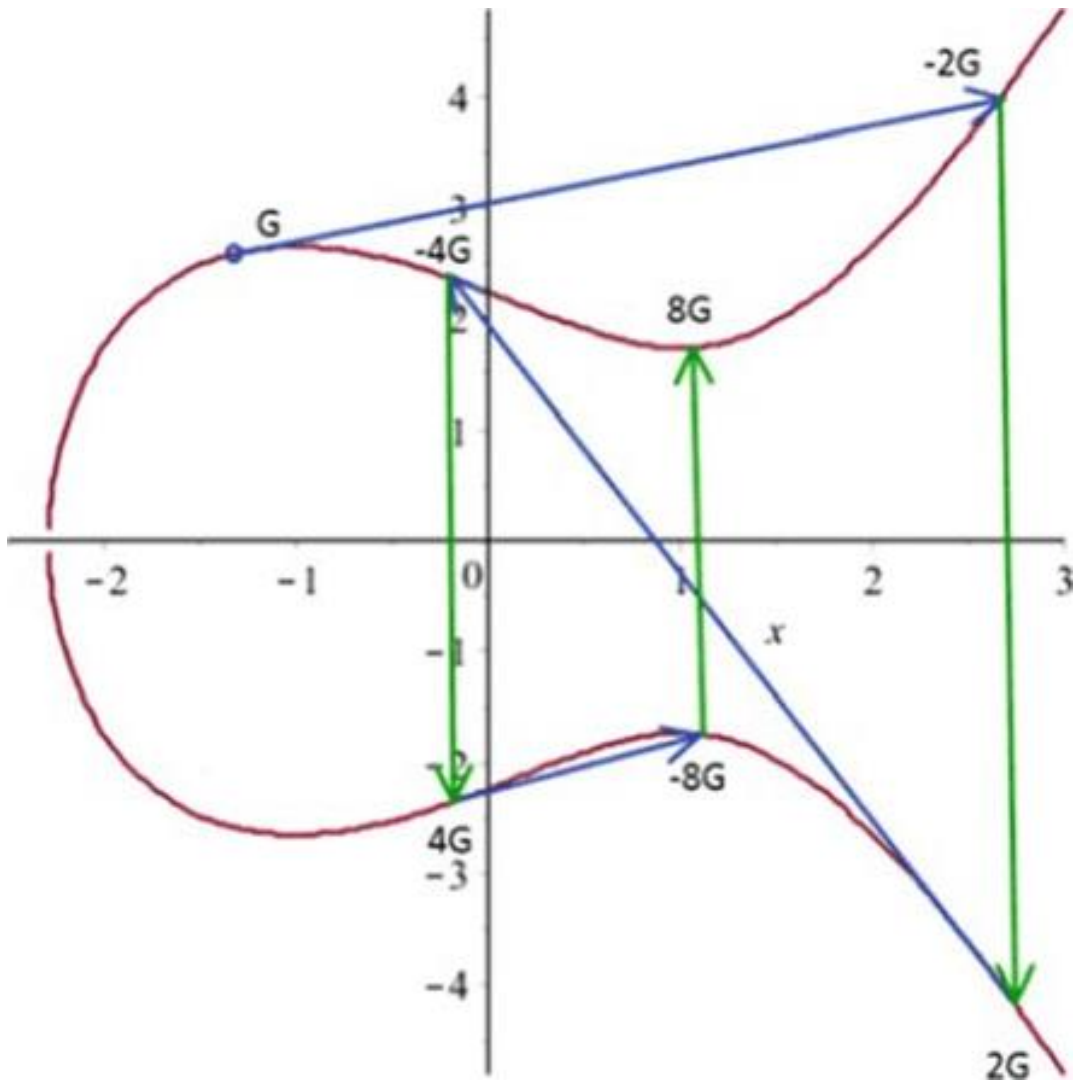


Elliptic Curve Cryptography-3

Module III

Adding P to P



Knowing P and the coordinates of $n \cdot P = (x, y)$ is a hard problem, If n is a very large number.

How to compute 2P by using algebraic expression?

- Let us consider the point $P(x, y)$ in the elliptic group $E_p(a, b)$.
- We have to consider $P(x_1, y_1) + P(x_2, y_2)$, where $x_1=x_2, y_1=y_2$.
- The rules for addition of P with P over the elliptic curve $E_p(a, b)$ is,
- $x_3 = \lambda^2 - x - x \pmod{p} \mid x_3 = \lambda^2 - x_1 - x_2 \pmod{p}$
- $y_3 = \lambda(x - x_3) - y \pmod{p} \mid y_3 = \lambda(x_1 - x_3) - y_1 \pmod{p}$
- $$\lambda = \frac{3x^2 + a}{2y} \pmod{p}$$
- E.g. $(4, 6), E_{11}(1, 1)$.
- $\lambda = \frac{3x^2 + a}{2y} \pmod{p} \rightarrow \frac{3 \cdot 4^2 + 1}{2 \cdot 6} \pmod{11} = \frac{49}{12} \pmod{11}$
- $49 \cdot 12^{-1} \pmod{11} = 49 \cdot 1 \pmod{11} = 5$.

How to compute 2P by using algebraic expression?

- $x_3 = 5^2 - 4 - 4 \pmod{11} \rightarrow 25 - 8 \pmod{11} \rightarrow 17 \pmod{11} \rightarrow 6$
- $y_3 = 5(4 - 6) - 6 \pmod{11}$
- $= -16 \pmod{11} = 11 - (16 \pmod{11}) = 11 - 5 = 6.$
- $2.P = (6,6) .$
- Check 2P lies on the curve, $y^2 \pmod{11} = x^3 + x + 1 \pmod{11}$
- $6^2 \pmod{11} = 6^3 + 6 + 1 \pmod{11}$
- $36 \pmod{11} = 223 \pmod{11}$
- $3 = 3$ ✓

Properties of elliptic curves

- Consider the elliptic curve E . The set of points on the curve over a finite field F_p , $E(F_p)$ is a finite abelian group. Let us begin by showing that $E(F_p)$ is a group.
- **Closure:** consider points $P, Q \in E(F_p)$ then $P+Q \in E(F_p)$. Hence closure holds.
- **Identity:** The Identity element is the point at Infinity.

$$P + O = O + P = P$$

- **Associativity:** Consider the points $P, Q, R \in E(F_p)$ then

$$(P+Q) + R = P+(Q+R)$$

- **Additive Inverses:** If $x_2=x_1$ and $y_2=-y_1$, that is $P=(x_1,y_1)$ and $Q=(x_2,y_2) = -P$, then $P - P = O$ (Point at Infinity). $P+Q=O$ ($Q=-P$). Hence, $E(F_p)$ is a group. $E(F_p)$ is abelian, for points $P, Q \in E(F_p)$, $P+Q = Q+P$. Hence, $E(F_p)$ is an abelian group.

Elliptic Curve Cryptography

- Steps:
- 1. Encode a plaintext message as point on the curve.
- 2. establish public private keys
- 3. Encrypt the message using the public key.
- 4. Decrypt using the private key.

Elliptic Curve Cryptography

- Steps:
- 1. Encode a plaintext message as point on the curve.

Let us consider $E_{11}(1,1)$

(4,6) directly take plain text message. There are many methods to convert plaintext to points on the elliptic curve.

Elliptic Curve Cryptography

- 2. Keys generation

Choose a generator point $G \in E_p(a, b)$

for example $(1,5) \in E_{11}(1,1)$

select a private key $n=2$

Compute public key $= 2 \cdot G = G+G = (1,5) + (1,5)$

$$\begin{aligned}\lambda &= \frac{3x^2+a}{2y} \bmod p = \frac{3 \cdot 1^2 + 1}{2 \cdot 5} \bmod 11 = \frac{4}{10} \bmod 11 \\ &= 4 \cdot 10^{-1} \bmod 11 = 4 \cdot 10 \bmod 11 \\ &= 40 \bmod 11 = 7.\end{aligned}$$

$$x_3 = 7^2 - 1 - 1 \bmod 11 = 49 - 2 \bmod 11 = 47 \bmod 11 = 3$$

$$y_3 = 7(1 - 3) - 5 \bmod 11 = 7(1 - 3) - 5 \bmod 11 = 3$$

Elliptic Curve Cryptography

- 3. Encrypt the message using the public key.
- $C = [(KG), (M+K.PU)]$, k is a random number.
- $C = [C1, C2]$
- $= [2. (1,5) , (4,6)+2.(3,3)]$
- $= [(1,5)+(1,5) , (4,6) + (3,3)+(3,3)]$
- $\lambda = 1, x_3 = 6, y_3 = 5.$
- $= [(1,5)+(1,5) , (4,6) + (6,5)]$
- $= [(3,3) , (4,5)]$

Elliptic Curve Cryptography

- 4. Decrypt the message using the private key.
- $M = C2 - [n \cdot C1]$
- $= (4,5) - [2 \cdot (3,3)]$
- $= (4,5) - [(3,3) + (3,3)]$
- $= (4,5) - (6,5)$
- $= (4,5) + -(6,5)$
- $= (4,5) + (6, -5)$
- $\lambda = 6, x_3 = 4, y_3 = 6.$
- $= (4,6)$

ECC

- ECC takes one sixth the computational effort to provide the same level of cryptographic security that you get with 1024-bit RSA and is 15 times faster.

Symmetric key sizes (bits)	RSA key sizes(bits)	ECC key sizes(bits)
80	1024	160
112	2048	224
128	3072	256
192	7680	384
256	15360	512